

CSMFAB000000000113-Vol-11

Scale 2 — Micro Amateur (500 g–5 kg): Complete Analysis

Series: CSMFAB000000000113 | **Volume:** Vol 11 of 29 | **June 2026 Notes:** Small boat operation; amateur rocketry class payload

1. Configuration Summary

Parameter	Value
Spear mass range	0.5–5 kg
Representative mass (m_rep)	2.8 kg
Launch depth	15 m
Vessel radius (R = 0.2d)	3 m
Rope length (L_rope)	15.3 m
Rope angle (alpha)	11.3 deg
Geometric efficiency	98.1%
Number of vessels (n)	4–6
Representative n	5

2. Velocity and Energy Budget

Stage 1 — Underwater Archimedean screw exit velocity:

$v_{\text{screw}} = 22 \text{ m/s}$ (saturated exit for spear at depth 15 m)

Stage 2 — Smart Rope delta-v:

$$\begin{aligned} \Delta v &= n * J_{\text{rope}} * \eta_v / m_{\text{rep}} \\ &= 5 * 1001 * 0.981 / 2.8 \\ &= 1785 \text{ m/s} \end{aligned}$$

Total exit velocity:

$$v_{\text{exit}} = v_{\text{screw}} + \Delta v = 22 + 1785 = 1807 \text{ m/s}$$

= **Mach 5.27** at sea level (343 m/s sound speed)

Kinetic energy at exit:

$$KE = 0.5 * 2.8 * 1807^2 = 4491.8 \text{ kJ}$$

3. Trajectory Analysis

3.1 Vacuum Ballistic (theoretical max, no atmosphere)

$$\begin{aligned} H_{\text{vacuum}} &= v_{\text{exit}}^2 * \sin^2(42 \text{ deg}) / (2g) \\ &= 1807^2 * 0.447 / 19.62 \\ &= 74.5 \text{ km} \end{aligned}$$

3.2 Atmospheric Trajectory (realistic, with drag)

$$\begin{aligned} H_{\text{atmosphere}} &= H_{\text{vacuum}} * 0.65 \text{ (empirical correction for atmospheric drag losses)} \\ &= 48.5 \text{ km} \end{aligned}$$

3.3 Maximum Range (flat Earth, optimal angle ~42 deg)

$$R_{\text{range}} = v_{\text{exit}}^2 * \sin(2*42 \text{ deg}) / g = 331.2 \text{ km}$$

4. Orbital Velocity Gap Analysis

LEO orbital velocity: 7,800 m/s Exit velocity achieved: 1807 m/s **Delta-V gap: 5993 m/s = 76.8% of orbital velocity remaining**

Second stage NOT needed for basic flight research

For orbital attempt from 48.5 km apogee: Required second stage delta-v:

$$\begin{aligned} dv_2 &= \sqrt{v_{\text{orbital}}^2 + 2*\mu*(1/r_{\text{apogee}} - 1/R_{\text{earth}})} - v_{\text{apogee}} \\ &= 7439 \text{ m/s (approximate; see Vol 17 for full analysis)} \end{aligned}$$

5. Vessel Specifications

Minimum vessel length: 5 m (must carry 15 m rope + crew + power) **Minimum vessel tonnage:** 1.0 tonnes (reaction during rope release) **Vessel displacement during launch:** 0.393 cm (acceptable) **Required sea state:** Beaufort 2 or below

6. Rope Specifications

Parameter	Value
Rope type	Smart Rope (CSMFAB000000000022 marine grade)
Segment size	25 mm cube
Segments per rope	611
Mass per rope	5.25 kg
Total rope system mass (n=5)	26.27 kg
Pre-tension force	5000 N
Release force (max, per rope)	50,040 N

7. Safety Margins

Component	Design Load	Material Limit	Safety Factor
Spear body (ZrB2-SiC)	4970 N · s / 0.0085 s	850 MPa	>50
UHMWPE rope fibers	50,040 N (max)	4250 MPa * 7.85e-7 m ² = 3,336 N/strand	2.0 (30 strands)
Rope attachment to spear	250200 N total	ZrO2 joint: 1300 MPa	>100

8. Practical Build Notes

Small boat operation; amateur rocketry class payload

Materials sourcing: - ZrB2-SiC spear: Treibacher Industrie AG (Austria), \$280-320/kg powder; flash-sinter in-house - UHMWPE fiber: DSM Dyneema SK99, \$35-55/kg; available in 100-500m spools - KNbO3-BaTiO3:

TDK 2025 lead-free catalog, \$180/kg in actuator form - Vessels: charter or own BFRP-hull vessels (available for offshore research)

Citations (Vol-11)

- CSMFAB000000000022 V2.0, Smart Rope specs, CSM 2026
- CSMFAB000000000109 V2.0, Archimedes screw exit velocity data, CSM 2026
- Vol 02 (this series), Smart Rope energy analysis
- Vol 05 (this series), Multi-vessel geometry
- Vol 06 (this series), Depth optimization data
- Bate, R.R., Mueller, D.D., White, J.E., Fundamentals of Astrodynamics, Dover 2020
- Anderson, J.D., Introduction to Flight, McGraw-Hill 2017
- SpinLaunch technical publications, 2024-2025

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